

Tutorial 2, Kinematics

Question 1:

The velocity of a particle is given by $\mathbf{v} = \{16t^2\mathbf{i} + 4t^3\mathbf{j} + (5t + 2)\mathbf{k}\}$ m/s, where t is in seconds. If the particle is at the origin when $t = 0$, determine the magnitude of the particle's acceleration when $t = 2$ s. Also, what is the x, y, z coordinate position of the particle at this instant?

Question 2

If the position of a particle is described according to *equation 1*

$$\mathbf{r} = [2\sin(2t)]\mathbf{i} + [2\cos(t)]\mathbf{j} + 2t^2\mathbf{k} \text{ m} \quad (1)$$

where, t denotes time in seconds and angles are in radians, what are the particle's velocity and acceleration after 3 s?

Question 3

The position of a ball can be described according to *equation 2*

$$\mathbf{r} = (3t^3 - 2t)\mathbf{i} - \left(4t^{\frac{1}{2}} + t\right)\mathbf{j} + (3t^2 - 2)\mathbf{k} \text{ m} \quad (2)$$

where, t represents time in seconds. Calculate the speed and magnitude of acceleration of the ball after 3 s.

Question 4

A ball moves along the circular path defined by *equation 3*.

$$x^2 + y^2 = r^2 \quad (3)$$

The y velocity component of the ball is given by $v_y = 2r\cos(2t)$. Derive the x and y acceleration components.

Question 5

A car travels along a helical path with its x, y and z axis positions defined by $d\sin\alpha t$, $d\cos\alpha t$, and $h - bt$, respectively. α, d, h and b are all constant. Determine the modulus of the car's velocity and acceleration.

Question 1

$$\mathbf{v} = 16t^2\mathbf{i} + 4t^3\mathbf{j} + (5t + 2)\mathbf{k} \text{ m.s}^{-1}$$

The acceleration is given by

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 32t\mathbf{i} + 12t^2\mathbf{j} + 5\mathbf{k} \text{ m.s}^{-2}$$

When $t = 2$ s

$$\mathbf{a} = 64\mathbf{i} + 48\mathbf{j} + 5\mathbf{k} \text{ m.s}^{-2}$$

The magnitude of the acceleration is given by

$$a = \sqrt{64^2 + 48^2 + 5^2} = 80 \text{ m.s}^{-2}$$

The displacement is given by

$$\mathbf{r} = \int \mathbf{v} dt = \left(\frac{16t^3}{3} + k_1\right)\mathbf{i} + (t^4 + k_2)\mathbf{j} + \left(\frac{5t^2}{2} + 2t + k_3\right)\mathbf{k} \text{ m}$$

Where k_1 , k_2 and k_3 are constants of integration. Since $\mathbf{r} = 0$ when $t = 0$,

$$k_1 = k_2 = k_3 = 0.$$

Then the displacement becomes

$$\mathbf{r} = \left(\frac{16t^3}{3}\right)\mathbf{i} + (t^4)\mathbf{j} + \left(\frac{5t^2}{2} + 2t\right)\mathbf{k} \text{ m.}$$

When $t = 2$ s

$$\mathbf{r} = \left(\frac{16 \times 8}{3}\right)\mathbf{i} + 16\mathbf{j} + \left(\frac{20}{2} + 4\right)\mathbf{k} \text{ m.}$$

So the x, y and z coordinates of the particle when $t = 2$ s are 43 m, 16 m and 14 m, respectively.

Question 2

$$\mathbf{r} = [2\sin(2t)]\mathbf{i} + [2\cos(t)]\mathbf{j} + 2t^2\mathbf{k} \text{ m}$$

where t represents time, in seconds. The velocity is given by

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = 4\cos(2t)\mathbf{i} - 2\sin(t)\mathbf{j} + 4t\mathbf{k} \text{ m.s}^{-1}$$

When $t = 3$ s

$$\mathbf{v} = 3.84\mathbf{i} - 0.28\mathbf{j} + 12\mathbf{k} \text{ m.s}^{-1}$$

The acceleration is given by

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = -8\sin(2t)\mathbf{i} - 2\cos(t)\mathbf{j} + 4\mathbf{k} \text{ m.s}^{-2}$$

When $t = 3$ s

$$\mathbf{a} = 2.24\mathbf{i} + 1.98\mathbf{j} + 4\mathbf{k} \text{ m.s}^{-2}$$

Make sure you haven't forgotten the units.

Question 3

$$\mathbf{r} = (3t^3 - 2t)\mathbf{i} - \left(4t^{\frac{1}{2}} + t\right)\mathbf{j} + (3t^2 - 2)\mathbf{k} \text{ m}$$

where t represents time, in seconds. The velocity is given by

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (9t^2 - 2)\mathbf{i} - \left(2t^{-\frac{1}{2}} + 1\right)\mathbf{j} + 6t\mathbf{k} \text{ m.s}^{-1}$$

When $t = 3$ s

$$\begin{aligned}\mathbf{v} &= (81 - 2)\mathbf{i} - \left(\frac{2}{\sqrt{3}} + 1\right)\mathbf{j} + 18\mathbf{k} \text{ m.s}^{-1} \\ &= 79\mathbf{i} - \left(\frac{2}{\sqrt{3}} + 1\right)\mathbf{j} + 18\mathbf{k} \text{ m.s}^{-1} \\ &= 79\mathbf{i} - 2.2\mathbf{j} + 18\mathbf{k} \text{ m.s}^{-1}\end{aligned}$$

The speed (modulus of $\mathbf{v}$) is given by

$$v = \sqrt{79^2 + 2.2^2 + 18^2} = 81 \text{ m.s}^{-1}$$

The acceleration is given by

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 18t\mathbf{i} + t^{-3/2}\mathbf{j} + 6\mathbf{k} \text{ m.s}^{-2}$$

When $t = 3$ s

$$\mathbf{a} = 54\mathbf{i} + \frac{1}{(\sqrt{3})^3}\mathbf{j} + 6\mathbf{k} \text{ m.s}^{-2}$$

The modulus of the acceleration is given by

$$a = \sqrt{54^2 + 0.19^2 + 6^2} = 54 \text{ m.s}^{-2}$$

Question 4

$$v_y = 2r\cos(2t).$$

So the corresponding displacement is given by

$$y = \int v_y dt = r\sin(2t) + k$$

where k is a constant of integration. If we define $t = 0$ when $y = 0$, $k = 0$. Then

$$y = r\sin(2t).$$

It is possible that some of you may have seen this result straight away, without going through all the steps.

Substituting this result into the equation

$$x^2 + y^2 = r^2$$

gives

$$x^2 + r^2\sin^2(2t) = r^2$$

$$\Rightarrow x^2 = r^2[1 - \sin^2(2t)] = r^2\cos^2(2t)$$

$$\Rightarrow x = r\cos(2t)$$

Then the x component of the velocity is given by

$$v_x = \frac{dx}{dt} = -2r\sin(2t)$$

And the x component of the acceleration is given by

$$a_x = \frac{dv_x}{dt} = -4r\cos(2t).$$

Similarly

$$a_y = \frac{dv_y}{dt} = -4r\sin(2t).$$

Question 5

The displacement of the car is

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (d\sin at)\mathbf{i} + (d\cos at)\mathbf{j} + (h - bt)\mathbf{k}.$$

The velocity of the car is given by

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (da\cos at)\mathbf{i} - (dasinat)\mathbf{j} - b\mathbf{k}.$$

The speed (magnitude of the velocity) is given by

$$v = \sqrt{(da\cos at)^2 + (dasinat)^2 + b^2} = \sqrt{d^2a^2 + b^2}$$

The acceleration of the car is given by

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (-da^2\sin at)\mathbf{i} - (da^2\cos at)\mathbf{j}.$$

The magnitude of the acceleration is given by

$$\left|\frac{dv}{dt}\right| = \sqrt{(d\alpha^2 \sin at)^2 + (d\alpha^2 \cos at)^2} = \sqrt{d^2 \alpha^4} = d\alpha^2.$$